MAHATMA GANDHIUNIVERSITY,KOTTAYAM

MGU-UGP(HONOURS)REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATIONNOVEMBER2025

FIRSTSEMESTER

Multi-DisciplinaryCourses(MDC)-MG1MDCFSQ101-BAKING TECHNOLOGY

(2024ADMISSIONONWARDS)

Duration:1Hours **MaximumMarks:35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PartA

MultipleChoiceQuestions

1. d). All of the above
2. a). Straight dough method
3. d). Cookies
4. c). Mixing ingredients in baking is very important
5. c). Butter
6. a). Production of carbon dioxide
7. b). Sodium Chloride
8. b). Unshortened cakes
9. b). Tortilla
10. c). Both A and B
11. a). granular sugar
12. d). Both A and B
13. d). cookie
14. a). brownie
15. b). Fats and Oils

PartB

ShortEssayTypeQuestions

Answerany**4**Questions

EachQuestioncarries**5**marks

16. Discuss the types of fat used in baking

- Butter – Adds rich flavor and aroma, helps in creaming, provides tenderness and good mouthfeel. Contains 80% fat and some moisture.
- Margarine – Economical substitute for butter, stable at higher temperatures, contributes to volume, texture and spread in baked goods.
- Shortening (Vanaspati / Hydrogenated fat) – Provides softness, flakiness and tenderness to pastries, cookies and cakes; has a high melting point for better aeration.
- Oil (Sunflower, Canola, Coconut oil) – Provides moisture, improves shelf life and makes products tender; used in muffins, carrot cakes and quick breads.
- Lard / Animal fat – Traditionally used in pie crusts and pastries; gives excellent flakiness, good shortening power and crisp texture.

17. Compare the different types of bread originated in various countries with examples

- French Breads – Characterized by hard crust and chewy crumb; made with simple ingredients. *Examples: Baguette, Brioche.*
- Italian Breads – Softer texture, olive oil enriched, often flavored with herbs. *Examples: Ciabatta, Focaccia.*
- German Breads – Dense and dark, often made with rye, whole grains and sourdough. *Examples: Pumpernickel, Volkornbrot.*
- Middle-Eastern Breads – Flatbreads, unleavened or lightly leavened; baked at high heat. *Examples: Pita, Lavash.*
- Mexican Breads – Mainly tortillas made from maize or wheat; soft and unleavened. *Example: Corn tortilla.*

18. Identify the steps in cake-making process

1. Preparation of ingredients & equipment – Measuring ingredients, lining tins, preheating oven.
2. Mixing / Creaming / Blending – Combining fat and sugar, adding eggs, incorporating dry ingredients.
3. Folding & Final Mixing – Ensuring uniform batter without overmixing; maintaining air cells.
4. Baking – Baking at correct temperature and time to ensure volume, texture and crumb structure.
5. Cooling & Finishing – Allowing the cake to cool, demolding, icing, frosting or decorating.

19. Explain the basic principles of baking

- Proper Ingredient Proportion – Correct ratio of flour, fat, sugar, eggs and liquid ensures balanced texture.
- Accurate Mixing Methods – Incorporating air, developing gluten and creating uniform batter/dough.
- Correct Baking Temperature – Ensures rise, setting of structure, browning and prevention of collapse.
- Heat Transfer – Conduction, convection and radiation change dough into a porous, light product.
- Moisture and Steam Formation – Steam expansion, gas production (CO_2 from leavening agents) help leavening and structure formation.

20. Compare the role of each ingredient in bread making

- Flour – Main structure-forming ingredient; gluten provides elasticity and strength.
- Yeast – Biological leavening agent; produces CO_2 for volume and light texture.
- Water / Liquids – Hydrates flour, activates gluten and yeast; essential for dough formation.
- Salt – Controls yeast activity, strengthens gluten and enhances flavor.
- Fats / Oil – Tenderizes crumb, improves softness, increases shelf life and volume.

21. Memorize the steps in bread-making

1. Measuring & Mixing – Combining flour, yeast, water, salt and other ingredients to form dough.
2. Kneading – Develops gluten structure, elasticity and gas-holding capacity.
3. Fermentation / Proofing – Dough rises due to yeast activity; improves flavor and texture.
4. Shaping & Final Proofing – Dough is shaped into loaves/rolls and allowed to rise again.
5. Baking & Cooling – Heat sets the structure, brown crust forms; cooling stabilizes bread texture.